Co-Citation Analysis on Data Mining and Random Forest: A Literature Review

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Abstract. This paper examines the intellectual network of research between the area of data mining and the Random Forest (RF) algorithms using the co-citation analysis. Based on 1,205 articles found in Web of Science, a co-citation map was created in order to determine the overall thematic clusters with relations. The resulting five distinct clusters found were along application-driven research areas: geospatial environmental modeling and risk assessment, landslide susceptibility prediction, machine learning toward hydrological and environmental systems, educational systems and behavior modeling, and educational data mining, including learning analytics. The analysis shows that environmental and geohazard research share a large overlap in methodology, whereas educational research is relatively isolated, regardless of sharing similar algorithms. The findings demonstrate the extent of the RF usage and the opportunity of method exchange across the disciplines. The conclusion of the paper includes an overview of the opportunities to increase cooperation across clusters, improve interpretability and optimization methods, and apply existing methods to new domains.

Keywords: Data Mining, Random Forests, Literature review, Machine Learning, Co-citation analysis, Environment Modeling, Educational Data Mining.

1 Introduction:

Random forest is an ensemble learning method, “a combination of tree predictors such that each tree depends on the values of a random vector sampled independently and with the same distribution for all trees in the forest.” [1]. The process of data mining has developed in such a way that it is a natural phase in the development of information technology. Initial advancements in data gathering and the development of databases proved instrumental towards the effective storing, retrieving, and processing of transactions on data. They are now common to most database systems, while “advanced data analysis (involving data warehousing and data mining) has naturally become the next step” [2].

It is necessary to mention that there are two different purposes to which RF can be applied in practice. In most applications, the focus is on developing “a classification or regression rule with good accuracy that is intended to be used as a prediction tool on

future data" [3]. Random Forests are also applied as feature selection, variable importance, and data relations analytical tools in other applications. Such adaptability has prompted the use of Random Forests in many different areas such as healthcare, education, finance, environmental science, and social sciences.

Co-citation analysis is a flexible method that can map the intellectual space of research domains by revealing patterns of shared citation. The co-citation relation arising when two papers are often co-cited in numerous studies indicates an intellectual connection and suggests that the two sources describe related issues, use similar or comparable methodological tools, or share similar theoretical orientations. Using co-citation in literature on the topic of data mining and Random Forests enables and informs works, and any new subjects that structure the thought of communities on the topic.

The paper has been organised with a view to providing a comprehensive treatment of the intellectual context of the analysis of data mining and Random Forests. Following this introduction, the paper reviews the findings of the prior research and identifies the key gaps in available knowledge on how the field is organized. The Data collection process is specified using Web of Science, followed by explaining how the co-citation network was created using the set of 1205 papers retrieved, and how the analytical methods were revised to help locate the clusters, and finally, to illustrate the entire map. The results section will present the findings of the co-citation network analysis, with the research clusters identified along with how the different themes of research connect. Finally, the discussion and conclusion sections interpret these findings, consider their implication in gaining an understanding of the intellectual progress of this area, and suggest future research lines.

2 Literature Reviews:

Lópes-Zambrano et al. [4] conducted a systematic literature review of 45 studies that used machine learning techniques, including Random Forests, Support Vector machines, and Decision Trees to predict academic success, risks of dropout, and success in students. The review highlights the usefulness of data obtained through the Learning Management Systems (LMS), assessments, and attendance data in predicting student learning outcomes. This literature is valuable to Cluster 5 (Educational Data Mining and Learning Analytics) as it provides information that can be used to understand ML-based prediction models and how to apply them in the education setting. There is also an overlap with Cluster 4 (Educational Systems and Behavior Modeling), and in this case, predictive models are used with student engagement data. Though methodological decisions fit time-series modeling in Cluster 3 quite sufficiently, they show no application to other non-educational contexts, thereby demonstrating an inability to cross-domain generalize. In addition, despite numerous studies focusing on prediction models, little has been done in terms of optimizing models to maximize long-term learning outcomes or incorporating interpretability frameworks into real-world education interventions.

Ordoñez-Avila et al. [5] in their review considered data mining methods applied in predicting teacher assessment in higher education using student data. The research analyzes the synthesis of 29 articles using soft computing methods like Fuzzy Neural Networks, Naive Bayes, RF, and SVM. The work is closely connected to Cluster 4, since it considers behavioral modeling of teachers by means of objective feedback processes such as sentiment analysis and appraisal forms. It is also highly correlated

with Cluster 5, especially where the student metrics are transformed to measure the effectiveness of instructions. Although introducing a comprehensive picture of algorithms and their applications, it is important to mention that, according to the review, many studies respectively ignore predictive modeling or skip the analysis of student performance as a direct input variable. This is an indication of a significant research gap: although the general idea of teacher evaluation is gaining a lot of popularity in the scientific circles, predictive modeling of teacher evaluation using fine-grained data of the students (such as grades, attendance rate, or course completion progress) remains unexplored in detail, particularly in high-stakes scenarios, such as online education or large educational establishments.

Hasan and Bao [6] provided a general overview of ML applications in medical informatics that incorporated more than 50 studies in which RF, Gradient Boosting Machines (GBM), and SVM were used to diagnose the disease, model treatment, and predict survival. The paper has a solid methodological connection to Cluster 3 (Machine Learning for Hydrological and Environmental Systems), despite being embedded in the medical field, because some of the methods conducted, such as missing data, model interpretability, and class imbalance, are also applicable to Cluster 3. It is also peripherally related to Clusters 1 and 4, which speaks to the generality of ML techniques as applied across domains. The review, however, recognizes the gap in the studies of domain-specific tuning and cross-contextual validation, particularly in cases of repurposing ML models across disciplines. This implies a research gap in transferring and cross-validating medical ML models to related high-dimensional non-linear issues in other systems, environmental (modeling student learning trajectories), or education (predicting landslides susceptibility based on environmental proxies).

Probst et al. [3] explored the process of fine-tuning hyperparameters in Random Forest models on various applications. They explored the effects of key tuning parameters and, in particular, the number of trees (ntree) and the size of feature subsets (mtry) on key performance measures such as accuracy and variance. Their study compared multiple optimization algorithms, such as grid search and Sequential Model-Based Optimization (SMBO), in areas including geospatial modeling and educational data mining. This in-depth technical review has a direct relation with the approach of Cluster 1 (Geospatial Environmental Modeling and Risk Assessment) and Cluster 2 (Landslide Susceptibility Prediction Models), where reaching high predictive integrity is essential. The results also have implications in Cluster 5, where properly tuned Random Forest models have proved to have better performances in predicting student academic outcomes. Nevertheless, the authors also point out a significant research limitation: although Random Forest parameter optimization has been developed and tested widely in environmental science contexts, there is residual ambiguity regarding its implementation and assessment within educational systems, especially in the setting of real-time learning. This is a gap that can be filled with future research into developing domain-specific hyperparameter optimization strategies that are applicable to Random Forest models in dynamic education settings.

3 Method and Dataset:

This paper uses a method called the co-citation, which involves examining scholarly articles based on the most common ones to be cited together. Researchers have been able to apply co-citation analysis to different fields such as machine learning, deep learning, and big data before expanding its application to diverse areas of research [e.g., 7,8].

The Web of Science database (www.webofknowledge.com) was used to obtain the data set, and a targeted search was conducted based on combining search terms "Data Mining" and "random forest", which are at the intersection of these two areas of research. This process gave the result of a large set of 1,205 papers, which can serve as a firm base for co-citation analysis. The resulting dataset reflects the wide range of application areas in which random forest methods are used in the context of data mining research.

After data collection, we downloaded the entire bibliographical entries in Web of Science in tab-delimited format, including the references cited in each article. Data preprocessing was carried out to ensure that all publications cited were assigned a unique identification scheme to facilitate accurate matching and avoid possible duplicates. This involved standardizing the names of the authors, matching issues of publications, and removing any irrelevant references to support the integrity of the analysis.

The bibliometric mapping tools were then used to perform co-citation analysis to create clusters of related publications. A co-citation network can be visualized to identify thematic groupings and intellectual connections throughout the dataset. These clusters indicate the interactions of some areas, including educational data mining, medical informatics, and environmental modeling, with Random Forest studies. The approach enables objective data-driven investigations in the field without the bias that can be introduced during manual literature review.

The choice to use only co-citation rather than including direct citation analysis is because we are interested in identifying structural relationships within the literature, which a simple popularity citation measure does not capture. Although the citation counts can be used to determine the most influential works, the patterns of co-citations can be used to understand the conceptual space of scholarship in greater detail by revealing which works scholars refer to most commonly in combination. The approach provides a more detailed picture of how the intellectual organization of these research communities operates in the field of random forest and data mining, with both established and new interdisciplinary relations between research groups.

4 Results:

The result of co-citation analysis is five different clusters in Figure 1. The nodes of the network indicate the publications, and the edges represent the co-citations. The node size represents the number of citations in the dataset, and the edge size indicates the co-citation value. There are three to eight nodes per cluster, with Cluster 1 being the largest. Most of the co-cited publications are those published after 2012, and there are also some previous seminal studies that refer to more recent advancements.

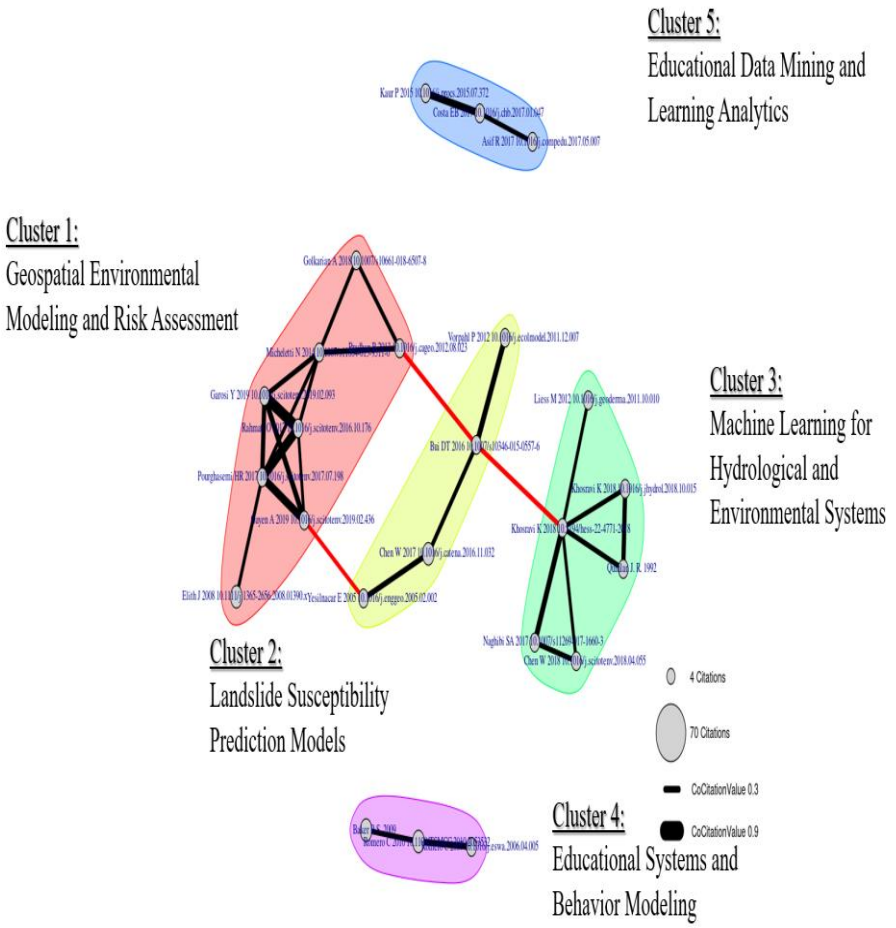


Fig. 2. Co-citation network of citations (nodes: number of citations >= 7 and edges: co-citation value >= 0.3)

As presented in Table 1, every cluster is associated with a particular application area. The five identified clusters have varied application areas, including geospatial environmental modeling and risk assessments, landslide susceptibility and prediction models, machine learning applications to hydrological and environmental systems, educational systems and learning behavior modeling, and educational data mining with learning analytics. All the clusters have a transparent emphasis on specific problems in the area where data mining methods and random forest algorithms are used to address practical issues. This trend shows that research in the area is quite application-oriented, and researchers are mainly concerned with solving practical issues in their relevant area of study rather than working out purely methodological solutions.

Table 1. Classification of Cluster topics

Cluster	Cluster topic	Category	Nodes
1	Geospatial Environmental Modeling and Risk Assessment	ML Application	8
2	Landslide Susceptibility Prediction Models	ML Application	4
3	Machine Learning for Hydrological and Environmental Systems	ML Application	6
4	Educational Systems and Behavior Modeling	ML Application	3
5	Educational Data Mining and Learning Analytics	ML Application	3

Cluster 1: Geospatial Environmental Modeling and Risk Assessment, points at the combination of spatial information and environmental modeling methodologies to assess ecological risks. Garosi et al. [9] offered machine learning technologies based on GIS to evaluate environmental hazards, and Rahmati et al. [10] established the importance of multiple geo-environmental factors in hazard mapping prediction. The thick red lines connecting this cluster with Cluster 2 suggest there is a substantial methodological intersection between environmental modeling research and landslide susceptibility studies.

Cluster 2: Landslide Susceptibility Prediction Models, focuses on the use of statistical and machine learning models to map landslides. Bui et al. [11] promoted hybrid methods that combine statistical analysis with ensemble learning methods, which increase predictive accuracy. Likewise, Chen et al. [12] showed the efficiency of more advanced algorithms like logistic model trees and random forests in landslide spatial prediction. The high co-citation connection with Cluster 1 indicates a robust methodological exchange between landslide hazard modeling and general geospatial risk assessment.

Cluster 3: Machine Learning in Hydrological and Environmental Systems corresponds to research that applies machine learning to hydrology and environmental monitoring. Chen et al. [13] also utilized ensemble methods to create predictive models of groundwater potential, demonstrating the utility of complex systems in the GIS-integrated environment. Naghibi et al. [14] further optimized the mapping of groundwater via support vector machines and random forests, as well as optimization techniques. The co-citation paths leading to Cluster 2 point to the fact that the techniques used in geohazard modeling are frequently borrowed by hydrological modeling.

Cluster 4: Educational Systems and Behavior Modeling deals with learning and modelling of the behaviour of students. One of the first systematic surveys in educational data mining was carried out by Romero and Ventura [15], which established the basis of adaptive learning applications. Following this, Baker and Yacef [16] surveyed the progress being made in behavioral indicators and predictive modeling to build more intelligent tutoring systems. Collectively, these publications informed the methodological foundation of the use of data mining in an educational context.

Lastly, Cluster 5: Educational Data Mining and Learning Analytics, extends the educational theme into large-scale performance prediction and learning analytics. Costa et al. [17] examined how to detect at-risk students in programming courses using academic records, whereas Kaur et al. [18] examined how to detect at-risk students in

general educational courses using classification methods. The shared purpose of enhancing student results via predictive modeling and learning analytics is evident in the associations within this cluster.

5 Discussion and Conclusion:

Co-citation analysis shows that the studies concerning data mining and Random Forests are highly application-oriented, and the related clusters are dedicated to environmental modeling, geohazard prediction, hydrological systems, and educational analytics. The clusters in environmental domains exhibit strong methodological connections, especially between geospatial modeling and landslide susceptibility, where the transfer of analytical methods between close areas is evident. The educational clusters, in contrast, are relatively more isolated, not evidencing as much methodological interaction even when identical algorithms were employed.

This division implies the existence of new possibilities related to cross-disciplinary knowledge exchange. As an example, existing hyperparameter tuning approaches used in environmental modeling might be repurposed to fit real-time learning analytics, and interpretability methodologies in medical informatics could even be used to increase trust and uptake in learning and environmental settings. Enhancing these relationships would enlarge the methodological aspects of every field.

To conclude, Random Forests are still a versatile and widely used approach, yet more interdisciplinarity may improve their efficiency. In the future, adaptation to new contexts should be tested on domain-specific adaptations, interpretability and optimization frameworks must be integrated, and future studies should promote methodological borrowing across what currently appear to be somewhat isolated sciences.

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